Chapter Two

Introduction HTML

TRUELINE
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Difference Between HTML,CSS and JS
HTML
HTML
+ CSS
HTML
+ CSS +
JS

HTML

- Hypertext Markup Language
  - is the standard markup language used to create web pages.
- HTML describes the structure of a website semantically along with cues for presentation
  - its a markup language, rather than a programming language
- *HTML documents consist of a tree of elements and text.*
- *Each element is denoted in the source by a start tag,*
  - *such as "<body>", and an end tag, such as "</body>".*
- Web browsers can read HTML files and render them into visible or audible web pages.

Anatomy of HTML document

```html
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="utf-8">
    <title>Hello World</title>
  </head>
  <body>
    <h1>Hello World</h1>
    <p>This is a web page.</p>
  </body>
</html>
```

- The <!DOCTYPE html> declaration defines this document to be HTML
- The <html> element is the root element of an HTML page
- The lang attribute defines the language of the document
- The <meta> element contains meta information about the document
- The charset attribute defines the character set used in the document
- The <title> element specifies a title for the document
- The <body> element contains the visible page content
- The <h1> element defines a large heading
- The <p> element defines a paragraph

HTML Anatomy ...

- Stylesheet (CSS)

```
<head>
  ...
  <link href="styles.css" rel="stylesheet">
<head>
```

- Javascript

```
<head>
  ...
  <script src="jscript.js"></script>
<head>
```

HTML Anatomy ...

- **Elements**
  - Elements define the structure and content of objects within a page.
  - Enclosed inside angel brackets < and > : E.g. <a> , , , , , <h1>…
  - meta information is placed in the head element
  - content information is placed in the body element
- **Tags**
  - The use of less-than and greater-than angle brackets surrounding an element creates what is known as a ***tag***.
  - Tags most commonly occur in pairs of opening and closing tags.
  - Content of the elements always falls between the opening (e.g.)and closing (e.g.) tag: E.g. …content here…
  - Tags have to be nested such that elements are all completely within each other, without overlapping:

Tag, Attribute and Value

- Tag
  - Void element: `<input> <hr> <br>`
  - Non void element:

    `<a>Link</a>`
- Attribute and Value

```
                Attribute
<p class="editor-note">My cat is very grumpy</p>
```

- Attributes
  - Attributes are properties used to provide additional information about an element.
  - The most common attributes include *id, class, src, herf*
  - Attributes are defined within the opening tag, after an element's name.

Tag, Attribute and Value

- Generally attributes include a name and a value.
- The format for these attributes consists of the attribute name followed by an equals sign and then a quoted attribute value.

- E.g.
  - **<a *href="demo.html"*> simple </a>**
  - **<input name="email" value="who@where.com">**
  - Atrribute: name, value
  - Value: email, who@where.com

Classifying HTML elements

- **top-level elements:** html, head and body
- **head** elements: placed inside the head
  - Title, style, link, meta, base, script
  - Wouldn't display on the page
- **block-level elements:** flow elements that behave like paragraphs
  - occupy 100% of available width
  - stacked vertically with preceding and subsequent block elements
  - article, h1-h6, header, footer, section, p, figure, canvas,
  - pre, div, ul, ol, dl, table, form, video

Classifying HTML elements

- **inline elements**
  - phrasing elements that behave like words
  - flow horizontally
  - usually placed inside block elements
  - a (anchor), audio, br, code, img, em, nav, samp
  - span, strong, sub, sup, time, var

Syntax of HTML

- All tags begin with < and end with >
- tag name is given immediately following opening <
- unrecognized tags and attributes are ignored
- attributes are given following the tag name:

  <tag attribute1="value" attribute2="value" ...>
- ' can be used instead of "
- forgetting to close a quote can result in a blank page
- It's good practice to keep HTML markup lowercase.
- values are case sensitive (used to).

Syntax of HTML

- elements without closing tags are of the form <img />
  - ending / is optional for HTML5, needed for polyglot documents
- elements must be well-formed: no bad nesting
  - Stuff
- attributes can be in any order.
- white space is allowed
  - between tag name and attributes
  - around the = sign
  - within attribute value (but should be avoided)

Standard HTML Tags

- Heading
  `<h1>, <h2>`, ... `<h6>`
- Table
  `<table>, <tr>, <th>, <td>, <tbody>, <thead>`
- Paragraph
  `<p>`
- List
  `<ol>, <ul>, <li>`
- Image
  `<img>`
- Horizontal Line
  `<hr>`
- Form
  `<form>`
- Form Component
  `<input>, <select>, <textarea>`
- Link/Anchor
  `<a>`
- Layer Box
  `<div>`
- Formatting
  `<strong>, <em>, <sub>`
- Line Break
  `<br>`

HTML 5

- HTML5 is a cooperation between the
  - World Wide Web Consortium (W3C) and
  - Web Hypertext Application Technology Working Group (WHATWG).

- WHATWG was working with web forms and applications,
- W3C was working with XHTML 2.0.
- In 2006, they decided to cooperate and create a new version of HTML.

HTML 5: Ground Rules

- Some rules for HTML5 were established:
  - New features should be based on HTML, CSS, DOM, and JavaScript
  - Reduce the need for external plugins
  - Better error handling
  - More markup to replace scripting
  - HTML5 should be device independent
  - Dev process should be visible to the public

HTML 5: New Features

- Reduce for external plugin
- Semantic elements: header, footer, section, article, others.
- canvas for drawing
  - paths of rectangles, arcs, lines, images
  - mouse events
- Better support for Local offline Storage (variation of cookies)
- audio & video elements
  - including drawing video on canvas
- form controls, like calendar, date, time, email, url, search, color, datetime, datetime-local, month, number, range, tel, week

New API's

- HTML Geolocation
- HTML Drag and Drop
- HTML Local Storage
- HTML Application Cache
- HTML Web Workers
- HTML SSE (Server-Sent Events)

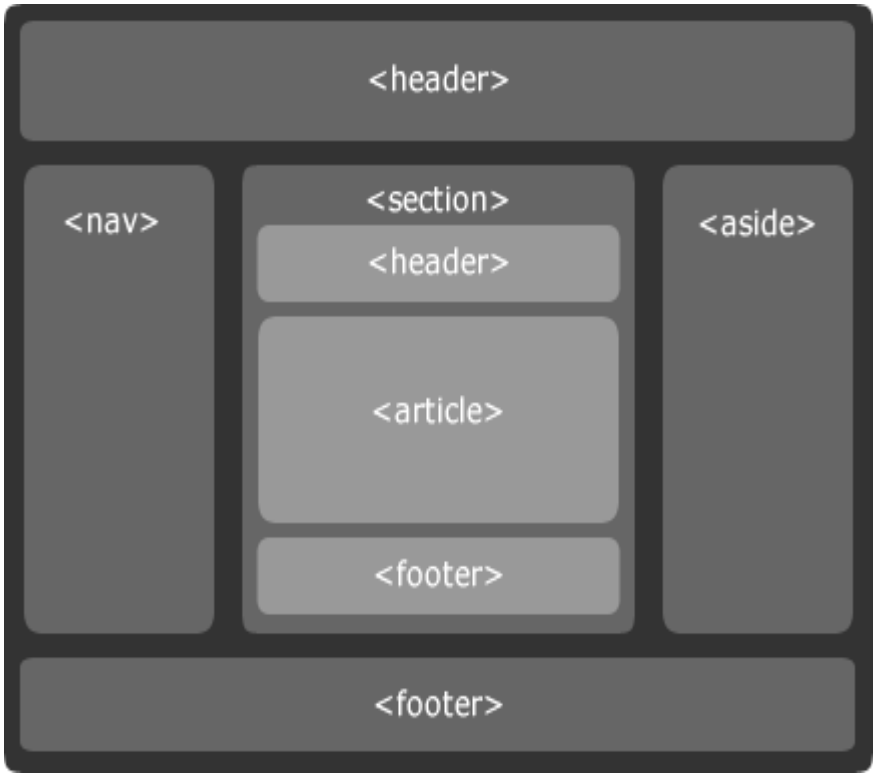
Cont…

- All HTML5 elements can have the following:
  - id – uniquely identifies an element in the page
  - style – gives presentation style
  - class – style class, or space-separated list of style classes
  - title – title for the element. Can be used as tool-top display
  - hidden – prevents element from being displayed when set to true
  - contenteditable, draggable, dropzone, spellcheck, and more

Cont…

- HTML5 defines eight new **semantic** elements.
- All these are **block-level** elements.
- To secure correct behavior in older browsers, you can set the CSS **display** property for these HTML elements to **block**:

```
header, section, footer, aside, nav, main,
article, figure {
  display: block;
}
```



HTML 5 migration

Typical HTML4	Typical HTML5
	<header>
	<nav>
	<section>
	<article>
	<footer>